

SHELL DEEPWATER SUBSEA GAS PIPELINE - ANNEX J QUALITY PLAN

Pipe Description: 28" (711.2 mm) x 17.50 mm, Grade X70MO PSL 2, LSAW (Offshore Subsea) | **Governing Standard:** Shell DEP 31.40.20.37 / API 5L Annex J / DNV-OS-F101

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.08%, P <= 0.012%, S <= 0.002%, CE_Pcm <= 0.19	Shell DEP / Annex J	H/W
2	Product Chemical Analysis	Two analyses per test unit	Pipe body coupon	ISO 14284	C <= 0.08%, P <= 0.012%, S <= 0.002%, CE_Pcm <= 0.19	API 5L Tablo J.1	W
3	Pipe Body Tensile Test	Once per test unit (lot <= 50 pipes)	Transverse body coupon	ASTM A370	Rt0.5: 485-605 MPa, Rm: 570-760 MPa, Af >= 19%, Y/T <= 0.90	API 5L Tablo J.2	H/W
4	CTOD Fracture Toughness Test	1 set (3 specimens) per qualification heat	Full thickness weld & HAZ	BS 7448 / ISO 12135	At -10 °C: Minimum CTOD δ >= 0.25 mm (No unstable brittle fracture)	Shell DEP / Annex J	H/W
5	CVN Charpy Impact Test (-20 °C)	1 set body + 1 set weld at -20 °C	Transverse Charpy V	ASTM A370	At -20 °C: Min Average 65 Joules, Min Individual 50 Joules	Shell DEP 31.40.20.37	H/W
6	DWTT (Dümen Aşırı Testi) at 0 °C	Once per heat at 0 °C	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear area >= 85%, No single specimen < 70%	API 5L Tablo J.2	H/W
7	Mill Hydrostatic Pressure Test	Each pipe (100% all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 100% SMYS (210 bar), Duration: MINIMUM 15 SECONDS	Shell DEP	H/W
8	Weld Seam 100% Automated UT + RT	100% longitudinal weld seam	Full weld seam	ISO 10893-11/6	100% Automated UT + Real-Time Radioscopy of full weld	Shell DEP / Annex J	H/W
9	Plate Body & Pipe Ends UT Laminar Scan	100% plate scan and 100% pipe ends	Full plate & pipe ends	ISO 10893-9/8	100% full plate UT (Class E1); 100 mm pipe ends (no defect > 4 mm)	API 5L Ek J & E	W
10	Strict Offshore Dimensional Tolerances	100% of all pipes (D, t, Ovality, L)	Pipe body & ends	API 5L Annex J.6	Diameter: +-0.30%; Out-of-roundness <= 0.50% D; Wall thickness: -5.0% / +10.0%	API 5L Tablo J.3	W